Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/04/2014
PI Name:	Patel, Zarana Ph.D.		
Project Title:	Development of an Autonomous, Dual Chamber Bioreactor for the Growth of 3-Dimensional Epithelial-Stromal Tissues in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	zarana.s.patel@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-3723
Organization Name:	KBRwyle/NASA Johnson Space Center		
PI Address 1:	Science, Technology and Engineering Group		
PI Address 2:	2400 NASA Parkway		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	NOTE: PI moved to Wyle in 2014; previously at Universities Space Research Association.		
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	10/01/2013	**End Date:**	08/11/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/11/2014 (originally 9/30/2014) because of PI move (Ed., 9/10/14)		
Key Personnel Changes/Previous PI:	ED. NOTE (9/10/2014): PI has changed institutions and work will continue as an internal NASA Johnson Space Center project. Title remains the same with appended note, "Development of an Autonomous, Dual Chamber Bioreactor for the Growth of 3-Dimensional Epithelial-Stromal Tissues in Microgravity--Wyle JSC." See that task for subsequent reporting.		
COI Name (Institution):	Huff, Janice (Universities Space Research Association)		
Grant/Contract No.:	NNX13AM07G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose the development of a novel, autonomous bioreactor that can provide for the growth and maintenance in microgravity of 3-D organotypic epithelial-stromal cultures that require an air-liquid interface. These complex 3-D tissue models accurately represent the morphological features, differentiation markers, and growth characteristics observed in normal human epithelial tissues, including the skin, esophagus, lung, breast, pancreas, colon. However, because of their precise and complex culture requirements, such as an air-liquid interface, these 3-D models have yet to be utilized for life sciences research aboard the International Space Station. The proposed technology will provide the capability to perform biological research on the ISS using these realistic, tissue-like human epithelial-stromal cell models and will contribute significantly to advances in fundamental space biology research on questions regarding microgravity effects on normal tissue development, aging, cancer and other disease processes. It will also allow for the study of how combined stressors, such as microgravity with radiation and nutritional deficiencies, affect multiple biological processes and provides a platform for conducting countermeasure investigations on the ISS without the use of animal models. The final design will be autonomous and consist of a dual chamber that provides for air-liquid, liquid-liquid, and liquid-air exchanges within the cell culture units. The bioreactor will support multiple tissue types and its modular design will provide for incorporation of add-on capabilities such as microfluidics drug delivery, media sampling, and in situ biomarker analysis. Finally, this proposal leverages the design of an existing 2nd generation prototype which has already been tested on parabolic flights through NASA's Flight Opportunities Program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This technology will provide the capability to perform biological research on the ISS using a realistic, tissue-like human epithelial-stromal cell model that accurately represents the morphological features, differentiation markers, and growth characteristics observed in normal human epithelial tissues. The enabling technology will contribute significantly to advances in fundamental space biology research on questions regarding microgravity effects on normal tissue development, aging, cancer and other disease processes.
Task Progress:	We have picked minimum design requirements for the mechanical parts and materials and researched available COTS options that would provide satisfactory performance with a small footprint. The modifications to the cell culture chamber were implemented into a CAD design model and we have manufactured several versions of the cell culture chamber and are conducting flow simulations in SolidWorks to determine areas of potential turbulence. Several trials on a breadboard prototype have been conducted. A biological test of the gas permeable membrane is currently being conducted as well. In the upcoming months, we will scale up the number of chambers on the prototype and evaluate pump parameters and microprocessor control with subsytems testing. We will also evaluate different designs of solenoid valves to assist with the scale-up. In addition, different programming languages will be evaluated as an option for building the GUI to coordinate with the microprocessor control. ED. NOTE (9/10/2014): PI changed institutions and work continues as an internal NASA Johnson Space Center project. Title remains the same with appended note, "Development of an Autonomous, Dual Chamber Bioreactor for the Growth of 3-Dimensional Epithelial-Stromal Tissues in Microgravity--Wyle JSC." See that task for subsequent reporting. Originally set up in July 2013 as new award at that time.
Bibliography Type:	Description: (Last Updated: 08/25/2020)